165365: foliated biotite granodiorite, Coomb Bore

Location and sampling

YOUANMI (SH 50-4), ATLEY (2741) MGA Zone 50, 745650E 6894870N

Sampled on 29 August 2001

The sample was taken from a 3 m-long boulder at the top of a 15 m-high rocky ridge, 1.5 km north of Coomb Bore.

Tectonic unit/relations

The sample is from a dark pinkish-grey, medium-grained porphyritic biotite granodiorite that has intruded the Edale Shear Zone of the Southern Cross Granite–Greenstone Terrane. The granodiorite contains abundant tabular feldspar phenocrysts, up to 5 mm long and aligned with the foliation. The foliation is subvertical and trends at 330°. The sample was collected to constrain the age of the Edale Shear Zone (Riganti, 2003).

Petrographic description

The principal minerals in this sample are plagioclase (45 vol.%), quartz (30 vol.%), orthoclase (20 vol.%), chloritized biotite (4 vol.%), opaque oxide (<1 vol.%), and apatite (<1 vol.%), with accessory altered possibly radioactive grains. This is an altered granodiorite with chlorite, sericite, and rare prehnite, as well as altered zircon. There are abundant rounded or euhedral feldspar augen, from 2 to 15 mm long in a foliated granitic matrix. The larger augen are clouded but untwinned orthoclase grains have rounded inclusions of quartz and sericite-clouded plagioclase. Smaller augen of plagioclase, irregularly dusted with sericite, are more abundant. The matrix is dominated by largely recrystallized quartz with a fabric in which the quartz c-axes are mostly within 30 to 40° of a foliation defined by chloritized biotite lamellae. Rare patches of prehnite are adjacent to the altered biotite. Some larger quartz grains are up to 1 mm in diameter, and there are lenses of fine-grained recrystallized plagioclase, generally clouded by clays. Lenses of opaque oxide and apatite are also present. Altered, possibly metamict grains up to 0.4 mm long could include altered zircon. These grains have been partly replaced by a reddish-brown or yellow-brown microcrystalline material.

Zircon morphology

The zircons isolated from this sample are typically colourless to dark reddish-brown and black, euhedral and equant to slightly elongate, and between 20 × 80 µm and 260 × 400 µm in size. Most grains have faint internal zonation or have unzoned, fractured interiors surrounded by strongly zoned exteriors, but a minority of grains are internally structureless. Many are metamict. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 30 June 2002. The counter deadtime during the analysis session was 24 ns. Twenty-one analyses of the CZ3 standard obtained during the analysis session indicated a Pb*/U calibration uncertainty of 3.48% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analysis 7.2, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Twenty-three analyses were obtained from 22 zircons. Results are given in Table 1 and shown on a concordia plot in Figure 2.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with both ancient and recent episodes of radiogenic-Pb loss. Eighteen concordant to slightly discordant analyses of 18 zircons have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 2686 ± 5 Ma (chi-squared = 1.08). Discordant analyses 4.1, 5.1, 6.1, 7.2, and 9.1 indicate slightly younger $^{207}Pb/^{206}Pb$ dates than the main population.

The date of 2686 ± 5 Ma indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of 18 concordant to slightly discordant analyses of 18 zircons is interpreted as the age of igneous crystallization of the granodiorite. The remaining discordant analyses are interpreted to be of analysis sites that have undergone radiogenic-Pb loss during several disturbance events.

Recommended reference for this publication:

NELSON, D. R., 2004, 165365: foliated biotite granodiorite, Coomb Bore; Geochronology dataset 218; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 30/06/2002; Data released: 06/12/2004

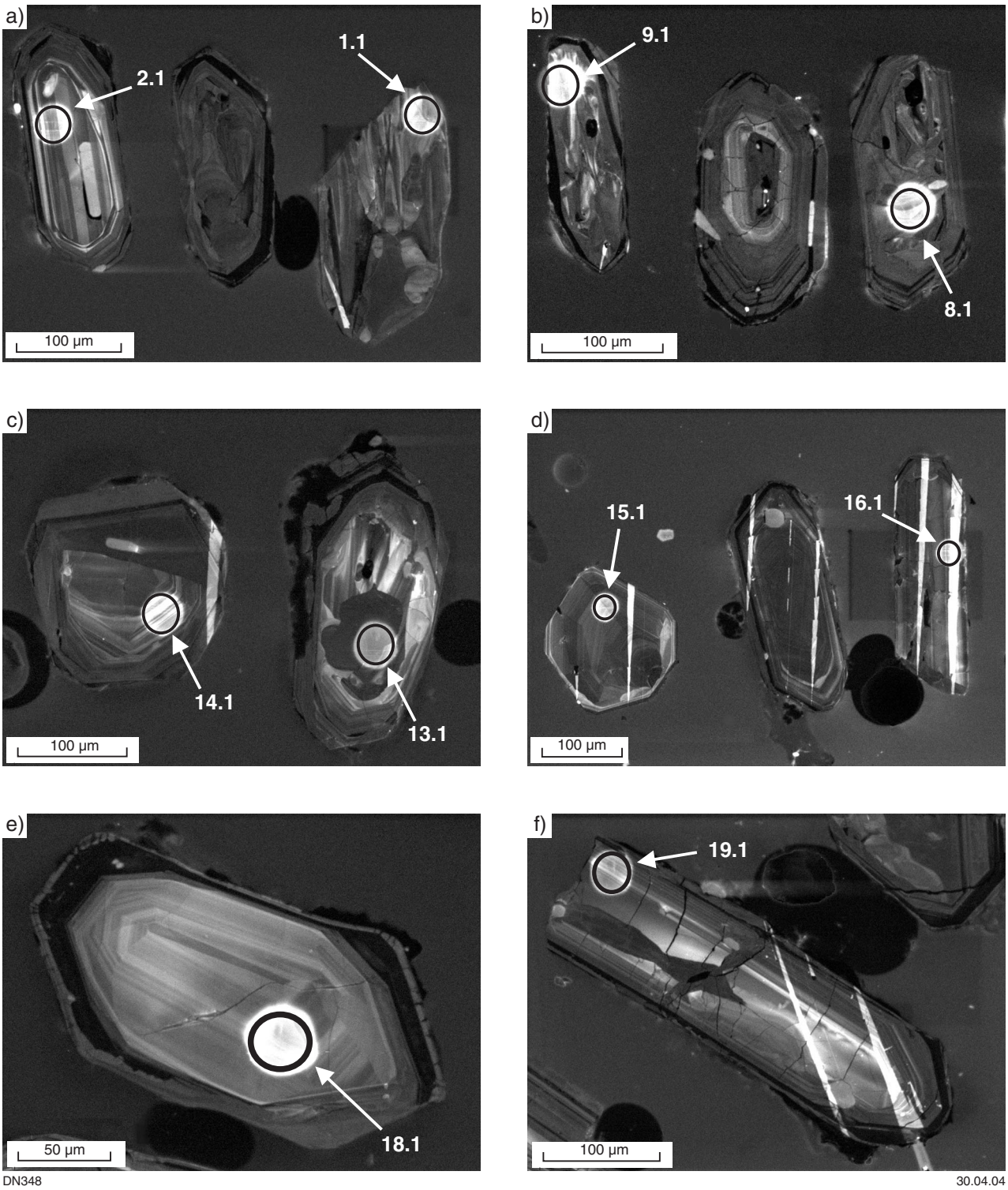

Figure 1. Cathodoluminescence images of representative zircons from sample 165365: foliated biotite granodiorite, Coomb Bore

Table 1. Ion microprobe analytical results for sample 165365: foliated biotite granodiorite, Coomb Bore

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	$\pm 1\sigma$	$^{208}Pb/^{206}Pb$	$\pm 1\sigma$	$^{206}Pb/^{238}U$	$\pm 1\sigma$	$^{207}Pb/^{235}U$	$\pm 1\sigma$	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	$\pm 1\sigma$
1.1	325	52	172	0.885	0.18412	0.00104	0.04411	0.00178	0.4858	0.0170	12.334	0.446	95	2 690	9
2.1	251	266	161	0.483	0.18373	0.00102	0.28872	0.00200	0.4991	0.0175	12.644	0.457	97	2 687	9
3.1	327	58	173	0.672	0.18527	0.00101	0.05379	0.00165	0.4834	0.0169	12.348	0.445	94	2 701	9
4.1	768	111	182	0.564	0.15374	0.00080	0.05180	0.00134	0.2242	0.0078	4.751	0.171	55	2 388	9
5.1	554	492	302	1.047	0.17214	0.00090	0.24565	0.00184	0.4321	0.0151	10.255	0.369	90	2 579	9
6.1	396	58	210	0.555	0.18083	0.00087	0.04149	0.00135	0.4955	0.0173	12.353	0.443	98	2 660	8
7.1	248	73	141	0.851	0.18327	0.00120	0.07628	0.00205	0.5087	0.0178	12.853	0.469	99	2 683	11
8.1	345	68	183	0.684	0.18403	0.00102	0.05614	0.00169	0.4858	0.0170	12.327	0.445	95	2 690	9
9.1	379	77	197	1.255	0.15305	0.00133	0.04832	0.00251	0.4802	0.0169	10.132	0.377	106	2 380	15
10.1	387	705	281	0.495	0.18517	0.00087	0.48427	0.00205	0.4968	0.0174	12.685	0.455	96	2 700	8
11.1	405	58	211	0.858	0.18289	0.00096	0.04473	0.00164	0.4790	0.0167	12.078	0.435	94	2 679	9
12.1	496	64	259	0.179	0.18286	0.00059	0.03459	0.00070	0.4953	0.0173	12.487	0.443	97	2 679	5
13.1	296	87	165	0.253	0.18426	0.00079	0.08245	0.00112	0.5051	0.0177	12.832	0.459	98	2 692	7
14.1	128	119	82	0.862	0.18229	0.00149	0.24984	0.00297	0.5027	0.0177	12.635	0.469	98	2 674	14
15.1	128	189	91	0.780	0.18514	0.00148	0.39670	0.00328	0.5098	0.0179	13.013	0.482	98	2 699	13
16.1	181	183	118	1.452	0.18214	0.00150	0.28017	0.00317	0.4983	0.0175	12.514	0.463	98	2 672	14
17.1	209	67	115	0.445	0.18321	0.00102	0.08587	0.00161	0.4947	0.0173	12.496	0.452	97	2 682	9
18.1	119	113	76	0.714	0.18194	0.00149	0.25609	0.00296	0.5038	0.0177	12.639	0.469	98	2 671	14
19.1	189	300	135	0.407	0.18348	0.00107	0.41988	0.00238	0.5122	0.0180	12.956	0.470	99	2 685	10
20.1	176	250	122	0.143	0.18480	0.00098	0.38743	0.00208	0.5103	0.0179	13.003	0.469	99	2 696	9
21.1	231	112	137	0.441	0.18421	0.00094	0.12994	0.00153	0.5152	0.0180	13.084	0.471	100	2 691	8
7.2	1 052	1 093	396	3.206	0.14079	0.00097	0.27207	0.00232	0.2828	0.0099	5.491	0.200	72	2 237	12
22.1	357	416	237	0.242	0.18218	0.00069	0.31541	0.00135	0.5117	0.0179	12.853	0.458	100	2 673	6

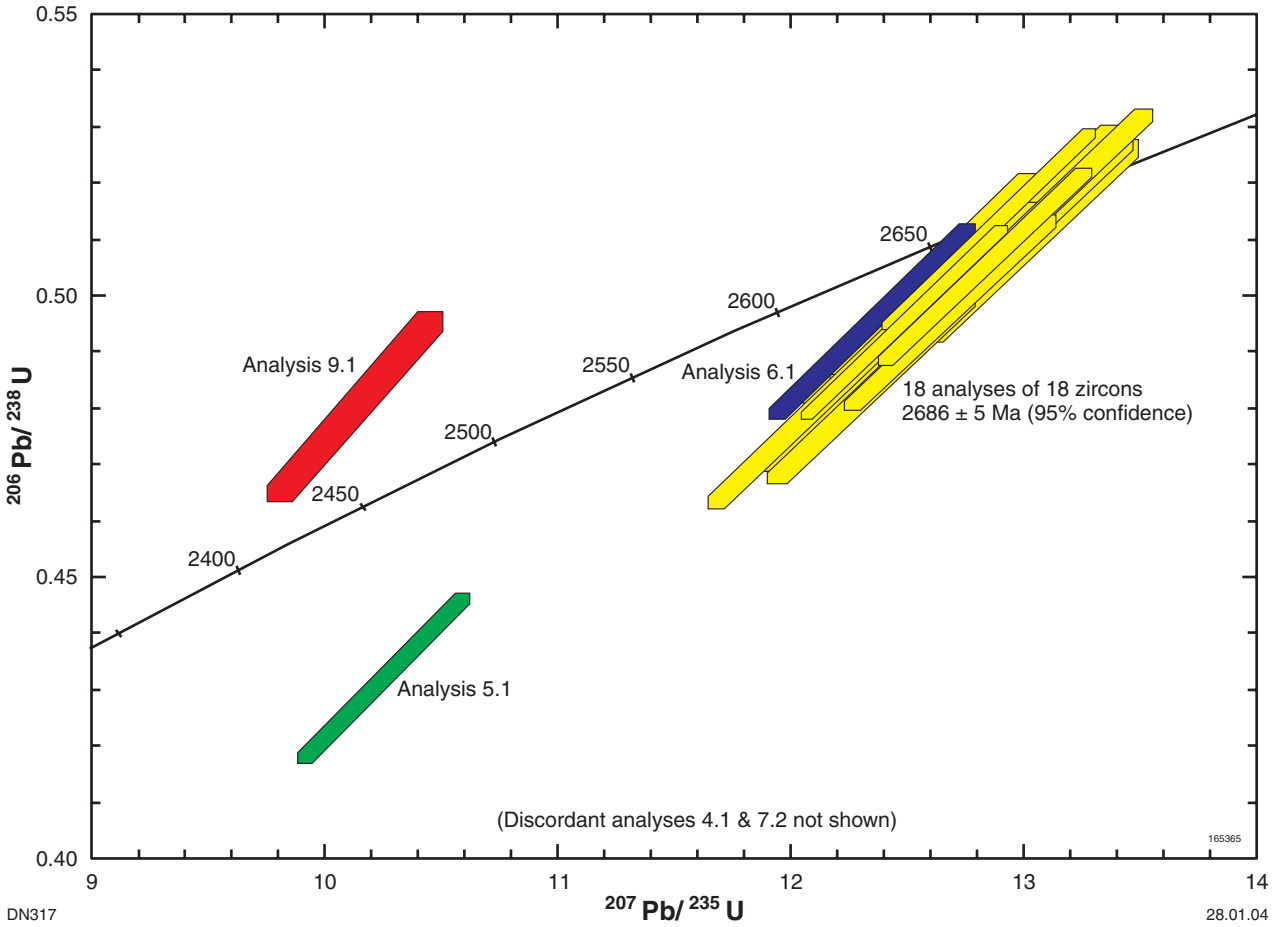

Figure 2. Concordia plot for sample 165365: foliated biotite granodiorite, Coomb Bore